CS204: Algorithms & Data Structures Lab

Week # 12 (1 Questions, 87 Marks)
Lab session: AL1
Held on: 21-Oct-2024 (Mon)
Lab Timings: 14:00 to 17:00 Hours   Pages: 2
Submission time: 16:45 Hrs, 21-Oct-2024
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Question 1: (87 points)

Templates and radix sort. You should submit the debugging log file with your name `roll_number.log`. Absence of log file leads to 0 marks for Task 03.

Task 01 (52 marks) Create a template class named `radix` with the following specification. Implement exceptions as necessary. Place `try`, `catch` blocks and `throw` atleast one exception.

1. (2 marks) Private members
   (a) A one dimensional array of typename T.
   (b) An integer representing size of array.
2. (50 marks) Implement the following public member functions
   (a) (2 marks) A constructor `radix` with the following specifications
      i. `Input argument 1` of type `typename T` one dimensional array
      ii. `Input argument 2` of type `int` base
      iii. `Functionality` Initialize the class members with the passed information.
   (b) (1 mark) A destructor with the following specifications
      i. `Input argument 1` None
      ii. `Functionality` Free class members memory
   (c) (20 marks) A member function `counting_sort` with the following specifications
      i. `Input argument 1` of type `int` representing digit position
      ii. `Return type void`
      iii. `Functionality` Performs sorting given numbers based on the digit position.
   (d) (20 marks) A member function `radix_sort` with the following specifications
      i. `Input argument 1 void`
      ii. `Return type void`
      iii. `Functionality` Perform sorting the array
   (e) (5 marks) A member function `get_max` with the following specifications
      i. `Input argument 1 void`
      ii. `Return type typename T`

iii. `Functionality` get the maximum number in the given array

(f) (2 marks) A member function `print` with the following specifications

i. `Input argument 1 void`

ii. `Return type void`

iii. `Functionality` Prints the elements of the array

Task 02 (15 marks) Perform the following in the main function. You should ignore all `delete` statement lines in the input file.

1. (5 marks) Read the input file `10-sort.txt`
2. Perform sorting on decimal inputs

3. (5 marks) Read the input file `08-sort.txt`
4. Perform sorting on octal inputs

5. (5 marks) Read the input file `16-sort.txt`
6. Perform sorting on hexadecimal inputs

Task 03 (20 marks) Perform the following for debugging

1. (1 mark) Compile the program to enable debugging
2. (1 mark) Invoke the executable using `gdb` debugger
3. (1 mark) Start logging debugger output messages
4. (1 mark) Set break point at first executable line in `main` function
5. (1 mark) Set break point at `radix_sort` function
6. (1 mark) Set break point at `counting_sort` function
7. (2 marks) Set condition break point that the count array at position 9 is 1.
8. (4 marks) proceed with debugging and demonstrate that count array at position reached 1.
9. (4 marks = 2 marks before; 2 marks after) Set `watch point` on the count array and demonstrate its value before and after watch point
10. (2 marks) Set `read watch point` on the base of the radix sort and demonstrate its use.
11. (2 marks) Set `catch point` on the exception generated in the (1 mark) `throw` statement and in the (1 mark) catch block.